

Progettazione di Applicazioni Mobili

Flutter

First steps

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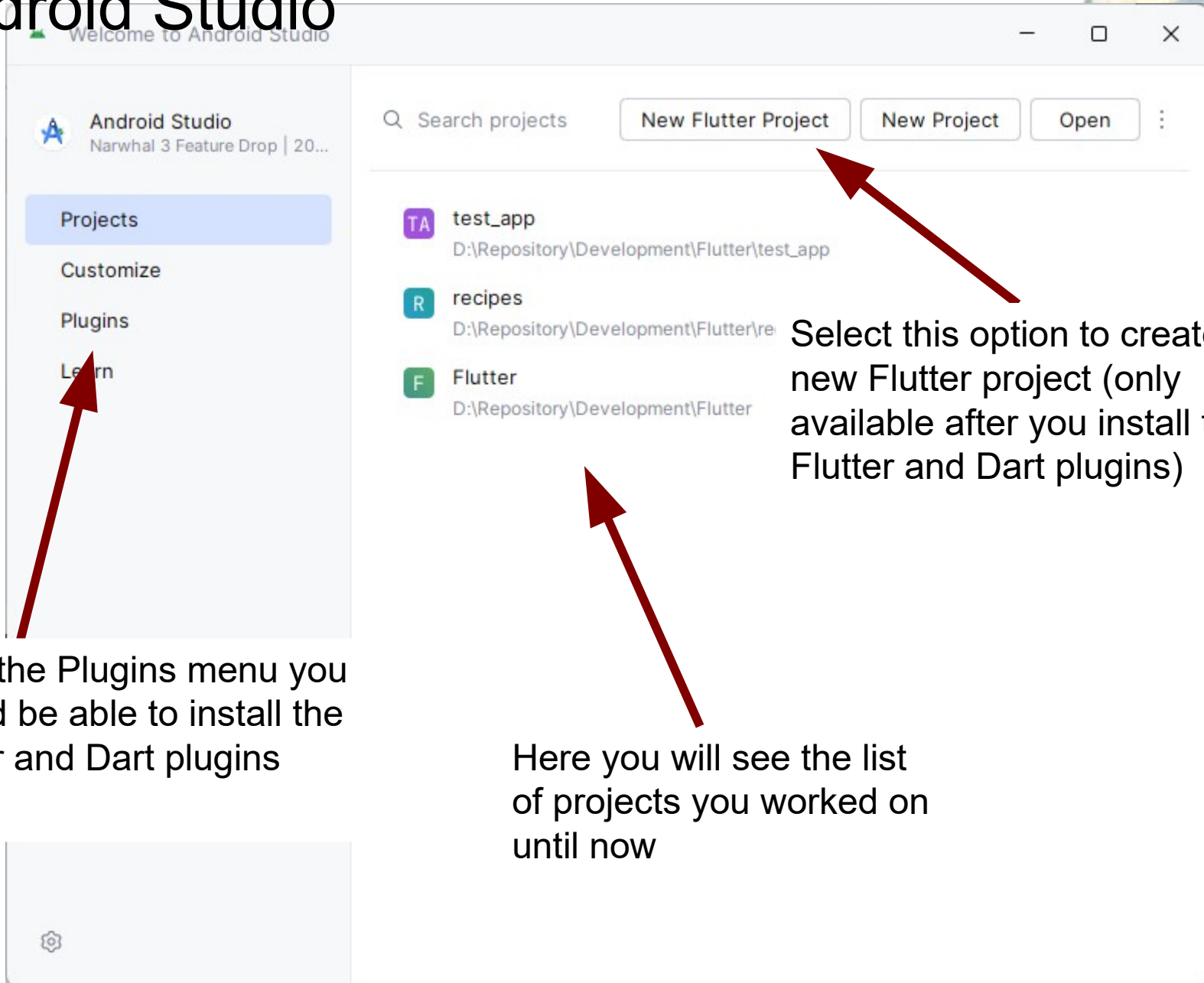


Flutter installation



- Step 1: download and install the Flutter SDK from the Flutter web site (<https://flutter.dev>), following the instructions for your operating system
- Step 2: download and install Android Studio from the Android Developers web site (<https://developer.android.com/studio>)
- Step 3: install the Flutter and Dart plugins for Android Studio from the Android Studio Welcome screen
- Set up the Android emulator (or your Android device) in Android Studio as explained in the Flutter web site

Android Studio



From the Plugins menu you should be able to install the Flutter and Dart plugins

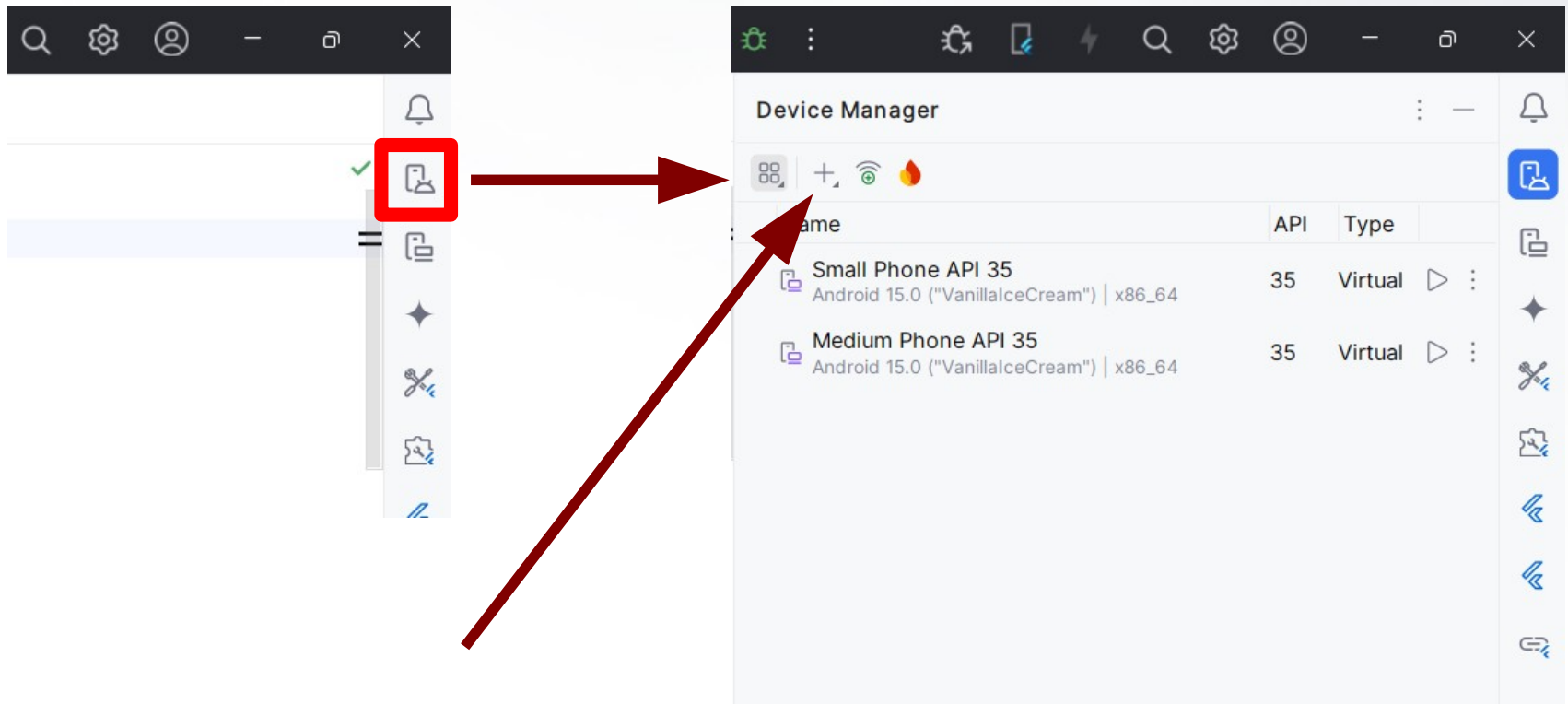
Select this option to create a new Flutter project (only available after you install the Flutter and Dart plugins)

Here you will see the list of projects you worked on until now

Set up the Android Emulator



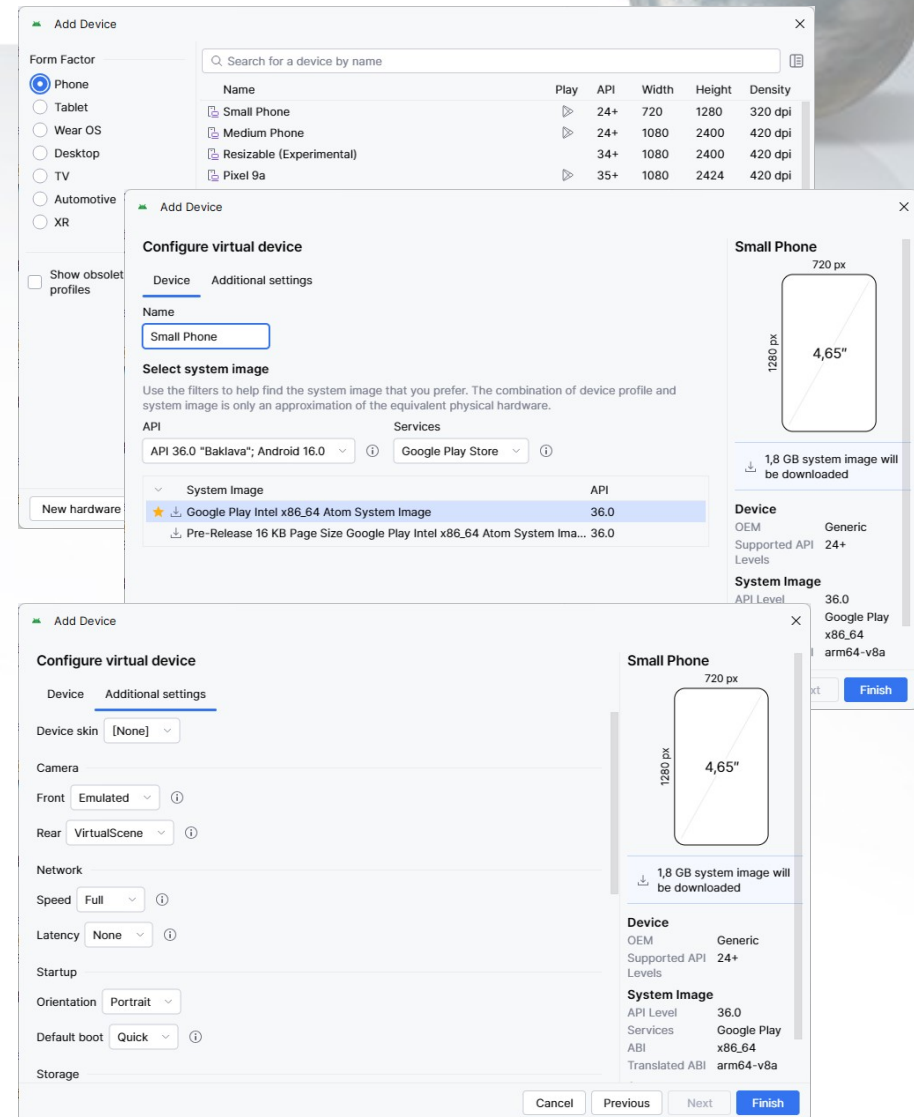
- To run applications on the Android emulator, it is necessary to create an Android Virtual Device



Creation of a new Virtual Device

Set up the Android Emulator

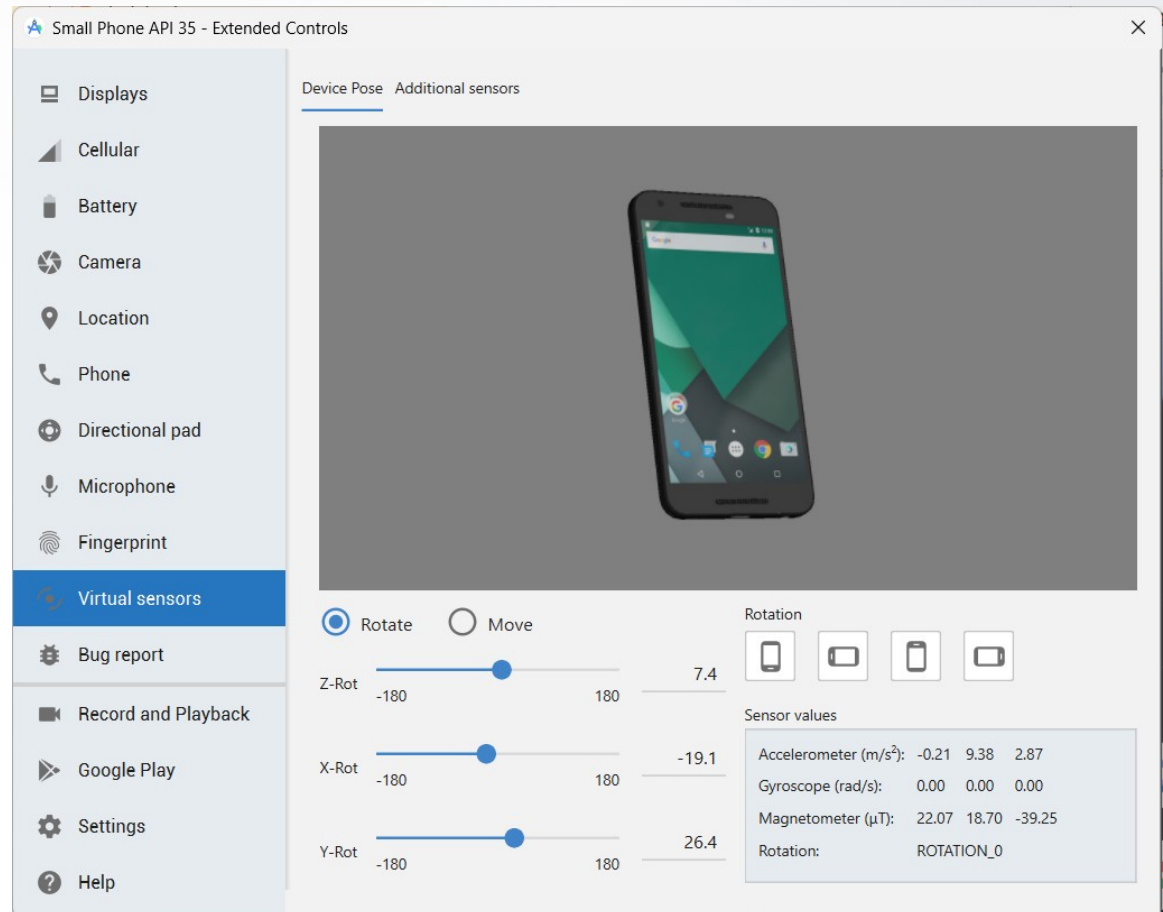
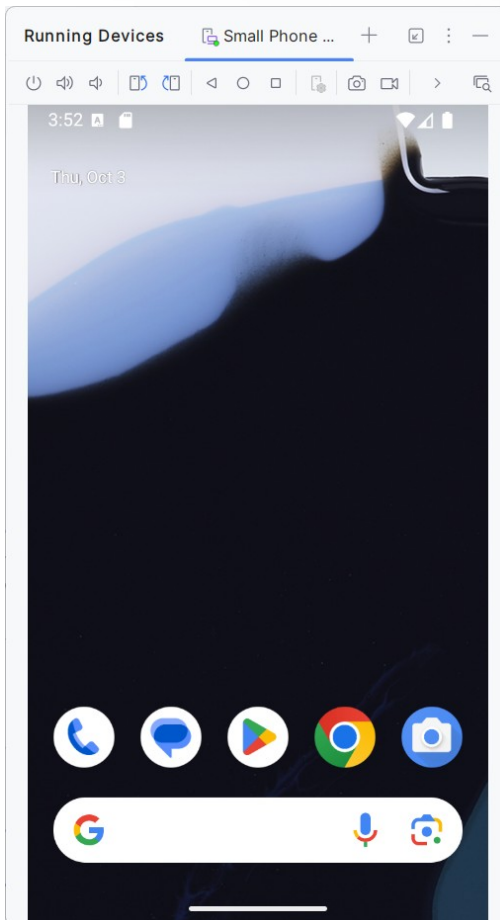
- A sequence of configuration screens allows you to choose:
 - Target device
 - Android version
 - Initial layout, camera, network, memory, performance options



Android Emulator



- The Android Emulator, included in the SDK, simulates most of the features of a real Android device



Android SDK Manager



- The Android SDK Manager makes it possible to download and update all needed Android platforms and tools. You can access it from the Configure menu or the Studio toolbar

Settings

Search

Appearance & Behavior

Keymap

Editor

Plugins

Version Control

Build, Execution, Deployment

Languages & Frameworks

C/C++

Android SDK

Dart

Flutter

JVM Logging

Kotlin

Markdown

Schemas and DTDs

Template Data Languages

Tools

Advanced Settings

Other Settings

Experimental

Languages & Frameworks > Android SDK

Manager for the Android SDK and Tools used by the IDE

Android SDK Location: C:\Users\Stefano\AppData\Local\Android\Sdk Edit Optimize disk space

SDK Platforms SDK Tools SDK Update Sites

Each Android SDK Platform package includes the Android platform and sources pertaining to an API level by default. Once installed, the IDE will automatically check for updates. Check "show package details" to display individual SDK components.

Name	API Level	Re...	Status
Android 15.0 ("VanillaIceCream")			
☐ Android SDK Platform 35	35	1	Not inst...
☐ Sources for Android 35	35	1	Not inst...
☐ Google APIs ARM 64 v8a System Image	35	8	Not inst...
☐ Google APIs Intel x86_64 Atom System Image	35	8	Not inst...
☐ Google Play ARM 64 v8a System Image	35	8	Not inst...
☒ Google Play Intel x86_64 Atom System Image	35	8	Installed
☐ Google Play Experimental 16k Page Size ARM 64 v8a : 35	35	3	Not inst...
☐ Google Play Experimental 16k Page Size Intel x86_64 : 35	35	3	Not inst...
☐ Pre-Release 16 KB Page Size Google APIs ARM 64 v8a : 35	35	3	Not inst...
☐ Pre-Release 16 KB Page Size Google APIs Intel x86_64 : 35	35	3	Not inst...
Android VanillaIceCream Preview			
☐ Android SDK Platform VanillaIceCream	VanillaIceCream	4	Not inst...
☐ Google APIs ARM 64 v8a System Image	VanillaIceCream	5	Not inst...

☒ Hide Obsolete Packages ☒ Show Package Details

OK Cancel Apply

Creating a Flutter project



- Select “New Flutter project” from the Welcome screen or through File > New > New Flutter Project... within Android Studio
- In the New Project screen specify the following data:
 - Project name
 - Project location
 - Type of Flutter application
 - Organization
 - Supported platforms

New Project

Project name:

Project location: ...

Description:

Project type:

Organization:

Android language: ☐ Java ☒ Kotlin

Platforms: ☒ Android ☒ iOS ☒ Linux ☒ MacOS ☒ Web ☒ Windows

When created, the new project will run on the selected platforms (others can be added later).

☐ Create project offline

More Settings

Module name:

Content root: ...

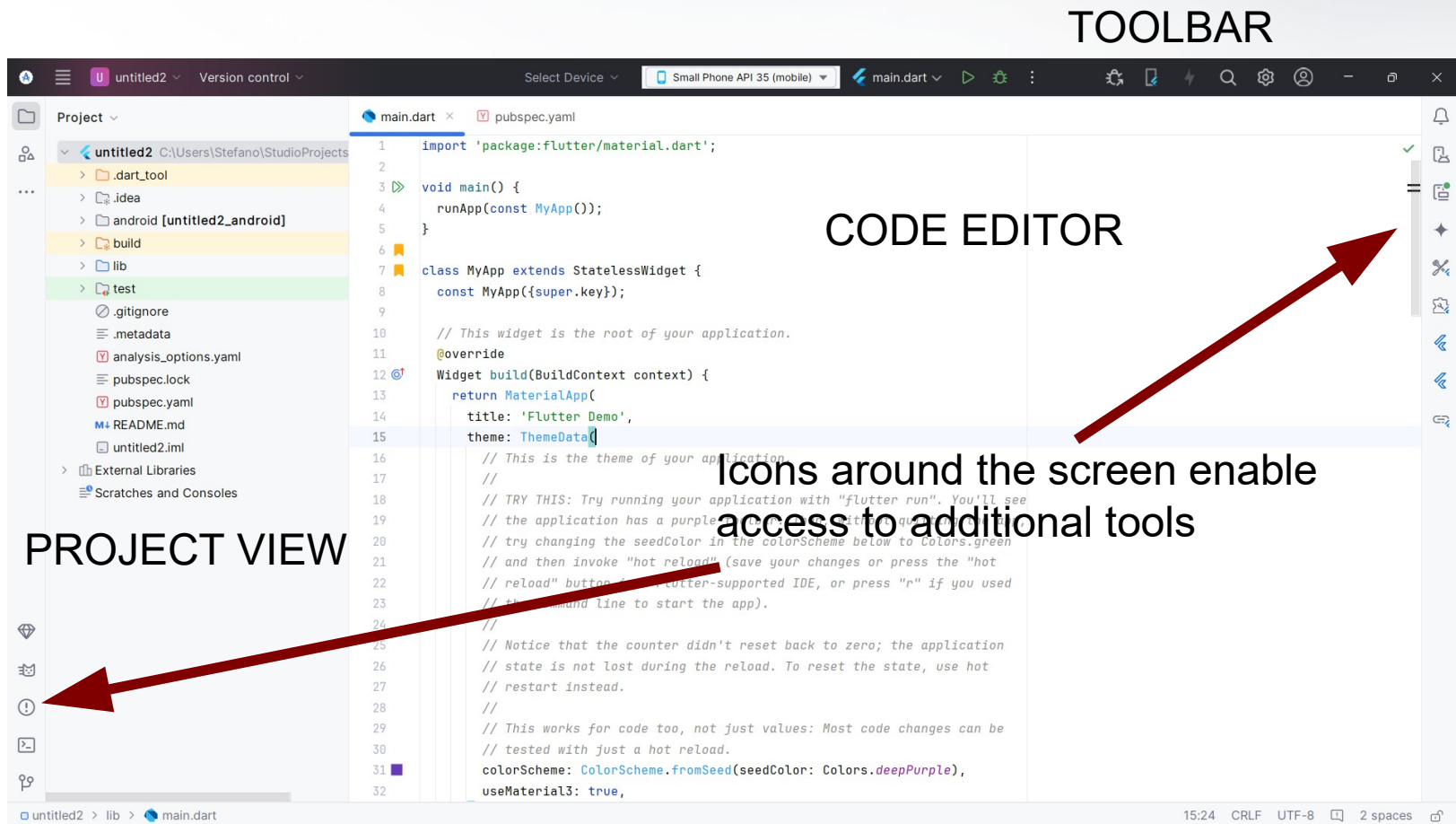
Module file location: ...

Project format:

Android Studio layout



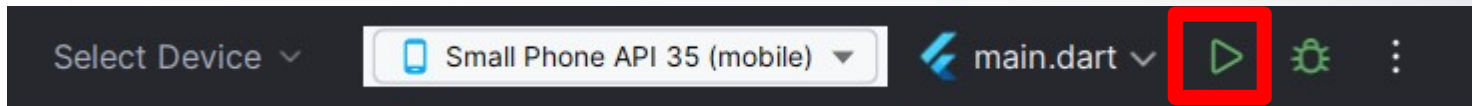
- Android Studio layout at the end of project creation (may change based on configuration)



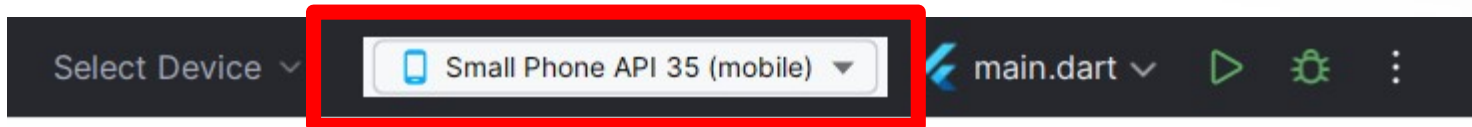
Running applications



- Applications can be started with the Run icon (or menu)



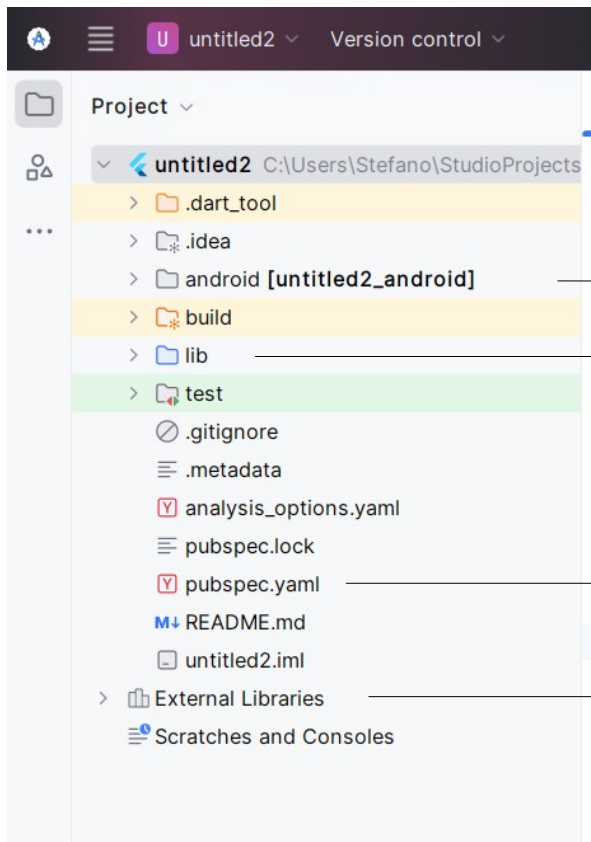
- Before running an application make sure you have an Emulator running and selected in the device dropdown



Flutter project structure



- The typical structure of a Flutter project includes several different files and directories



Android/iOS project directories

main Dart code

project configuration file

libraries

Basic structure of the default app



- Every Flutter app's main entry point is the *main()* method
- The *main()* method just includes a call to the *runApp()* method, provided by Flutter itself, and passes to it the top-level widget *MyApp*
 - There is always a single widget at the top of the hierarchy that contains all the others
- The *MyApp* class is a stateless widget so the only thing it must provide is a *build()* method
 - The widget returned from *build()* is an instance of *MaterialApp*, which provides the basic framework for a Material (i.e., Google UI-style) app

Basic structure of the default app



- The MaterialApp widget has a single child, which is an instance of the MyHomePage class
- The MyHomePage class defines a stateful widget, so in this case, we need two classes, the “core” class that extends from StatefulWidget and the state class associated with it that extends State
 - it’s the state class, _MyHomePageState, that effectively is the widget
- The *build()* method of _MyHomePageState returns a single Scaffold widget, which provides the fundamental visual layout for the app, including hooks for the App bar and the FAB

Basic structure of the default app



- The body argument passed to the Scaffold's constructor is how we add other widgets as children of this one
 - we have a Center widget, which is a container widget that centers its one and only child, a Column widget that lays out its children in a column
 - This Column has two children, both Text widgets, one for the static "You have pushed the button this many times:" text, and the other to display the number of times you've pushed the button